ARGUS ATTENDANCE SYSTEM

COMPLETE ENTERPRISE FEATURE GUIDE & OPERATING MANUAL

Next-Generation AI Biometrics, GPS Geofencing, Automated Payroll & Multi-Branch Operations

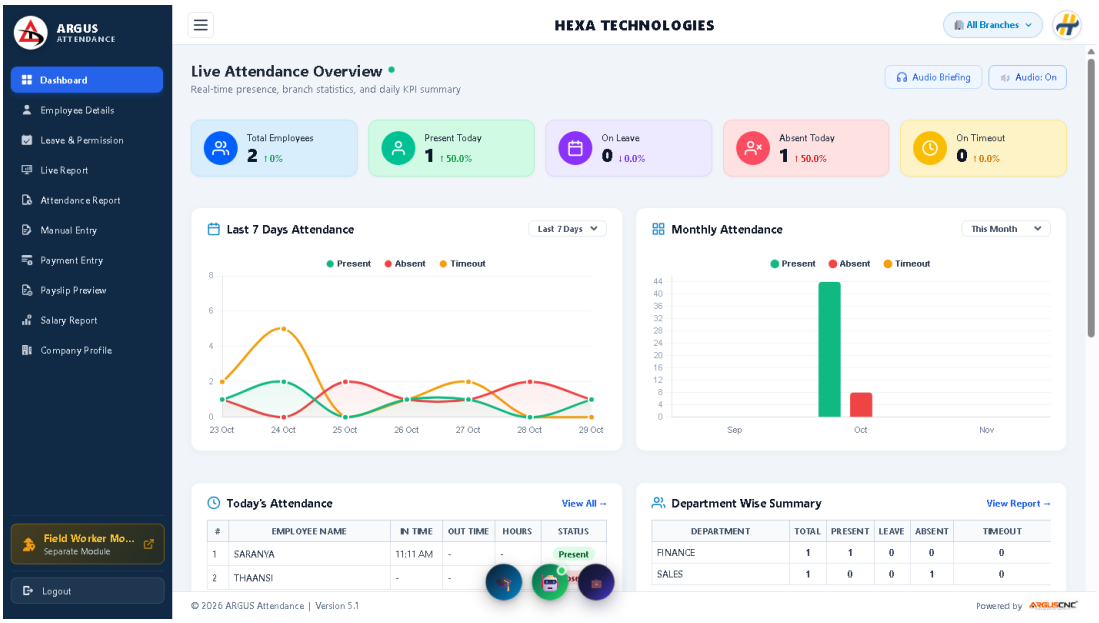


Figure 0: ARGUS Attendance Executive Operations Dashboard

Document Version:	Version 5.1 (Production Enterprise Release)
Target Audience:	Corporate HR Directors, System Administrators, Operations Executives
Core Platform:	Flask / Python 3.12, MongoDB Atlas, Deep Learning Face Engine, Leaflet GPS
Security & Compliance:	PBKDF2/SHA-256 Hashing, GPS Radius Locking, Anti-Spoofing Liveness AI
Publisher:	ARGUS TECHNOLOGIES (Coimbatore, Tamil Nadu, India)
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EXECUTIVE SUMMARY & PLATFORM ARCHITECTURE

The **ARGUS AI Attendance System** is a multi-tenant enterprise human capital platform engineered to automate staff identification, attendance logging, leave governance, and end-to-end payroll computation without requiring expensive biometric hardware. Utilizing computer vision neural networks, the system converts standard webcams and smartphone cameras into sub-second facial recognition terminals with anti-spoofing liveness verification.

Key operational advantages of the platform include:

- **Zero Proprietary Hardware:** Deploy instantly on laptops, tablets, or mobile phones without expensive finger scanner maintenance.
- **GPS Geofence Locking:** Enforces strict circular coordinate boundaries (100m to 500m) to guarantee that attendance can only be punched within verified workplace premises.
- **Single-Click Payroll & Payslips:** Automatically calculates normal work hours, late mark deductions, approved leaves, and overtime hours to generate statutory salary slips.
- **Multi-Branch Synchronization:** Consolidates data from headquarters, factories, branch offices, and field teams into a unified real-time dashboard.
- **Mobile Field Worker Tracking:** Delivers GPS check-in logging, customer visit verification, proof photos, and automated travel distance & petrol allowance accounting.

Component	Technology / Implementation	Function & Advantage
AI Face Engine	128D Deep Metric Learning	Sub-second face matching with ambient light invariance
Cloud Database	MongoDB Atlas Clustered Replica	Real-time distributed JSON records with indexing
Geofencing	Haversine Distance Geometry	Locks physical punches within exact workplace meters
Payroll Engine	ReportLab Vector Generation	Generates pixel-perfect corporate payslips & payroll registers
Field Tracker	OpenStreetMap / Leaflet Engine	Live route tracing, customer visits & petrol reimbursement

DOCUMENT STRUCTURE & TABLE OF CONTENTS

Section 1: Multi-Tier Authentication & Access Portals (Company, System Admin, Employee, Field)

Section 2: Executive Operations Dashboard & Live KPI Metrics

Section 3: AI Face Recognition Attendance Kiosk Terminal

Section 4: Employee Directory & Biometric Enrollment Management

Section 5: Real-Time Monitoring, Analytics & Audit Overrides (Live Stream, Reports, Manual Entry)

Section 6: Leave Governance, Permissions & Consolidated Balance Sheet

Section 7: Automated Payroll, Salary Advances, Disbursals & Monthly Payslips

Section 8: Enterprise Configuration, Shift Timings & GPS Geofencing Parameters

Section 9: Multi-Branch Network Management & Cross-Location Switching

Section 10: Mobile Field Worker GPS Tracking & Travel Allowance System

Section 11: Customer Support Help Desk & Super Admin Fleet Administration

Section 12: Security Specifications, Data Integrity & Technical Support

SECTION 1: MULTI-TIER AUTHENTICATION & ACCESS PORTALS

Security in the ARGUS ecosystem is built upon role-segregated gateways. Each user tier (Company Administrators, Super Administrators, Field Staff, and General Employees) accesses a dedicated authentication portal designed specifically for their operational context.

1.1 Company Administration Login Gateway

The primary enterprise portal for managing day-to-day operations. Company Admins log in using their authorized corporate email and encrypted credentials, or via Google Workspace OAuth single sign-on. The system automatically initializes tenant-scoped session tokens, loads geofencing policies, and syncs branch assignments.

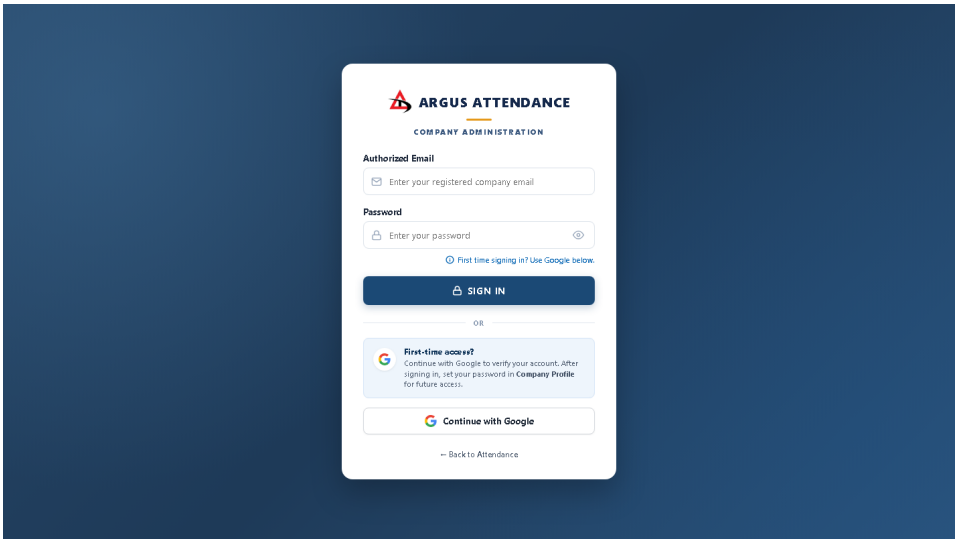


Figure 1.1: Company Administration Login Portal with Google OAuth integration

1.2 System Administration (Super Admin) Gateway

Super administrators manage tenant provisioning, subscription quotas, and platform-level configurations across all client organizations through an isolated security barrier.

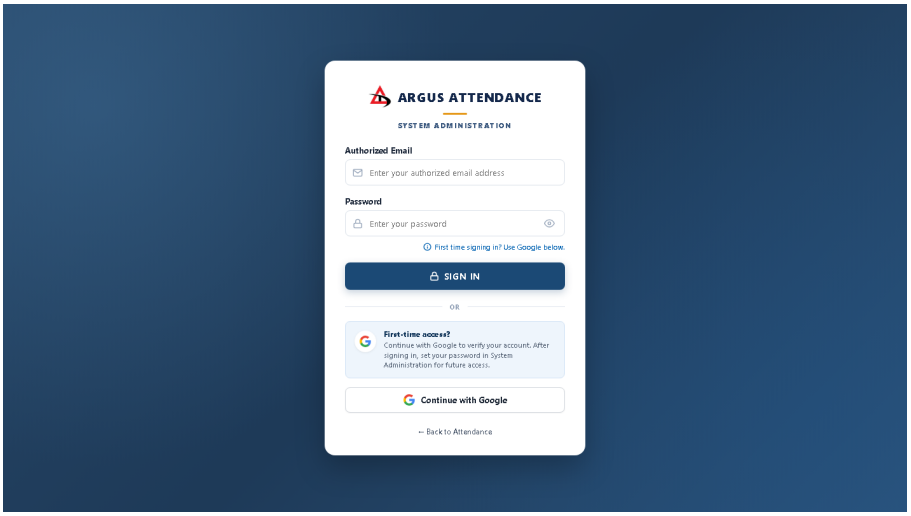


Figure 1.2: Super Administration Access Gateway

1.3 Employee Self-Service & Field Worker Mobile Portals

To empower mobile and remote employees, specialized responsive portals are provided for viewing personal attendance records, submitting leave applications, downloading payslips, and logging customer visits.

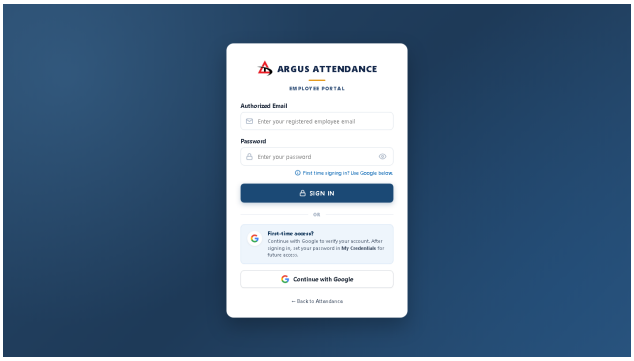


Figure 1.3: Employee Self-Service Login

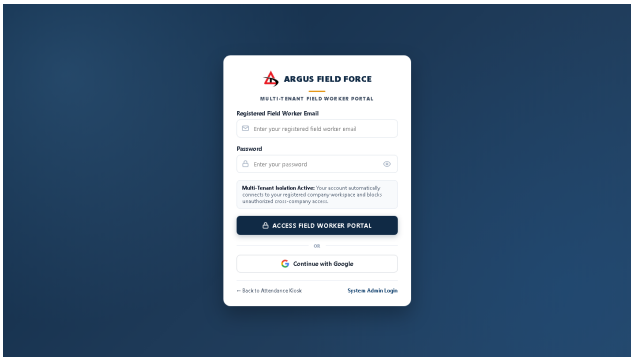


Figure 1.4: Field Worker Dedicated Mobile Login

■ Enterprise Authentication Highlights

- **Password Security:** Cryptographically salted PBKDF2/SHA-256 hashes prevent brute force attacks.
- **Persistent Session Tokens:** Unbreakable multi-tab remember tokens survive browser restarts safely.
- **Role Segregation:** Strict middleware guarantees employees cannot escalate privileges to administrative routes.

SECTION 2: EXECUTIVE DASHBOARD & LIVE OPERATIONAL KPIS

The central ARGUS Dashboard provides operational leaders with an instant, real-time command center. Built with asynchronous data polling, it delivers immediate visibility into workforce readiness without manual page refreshing.

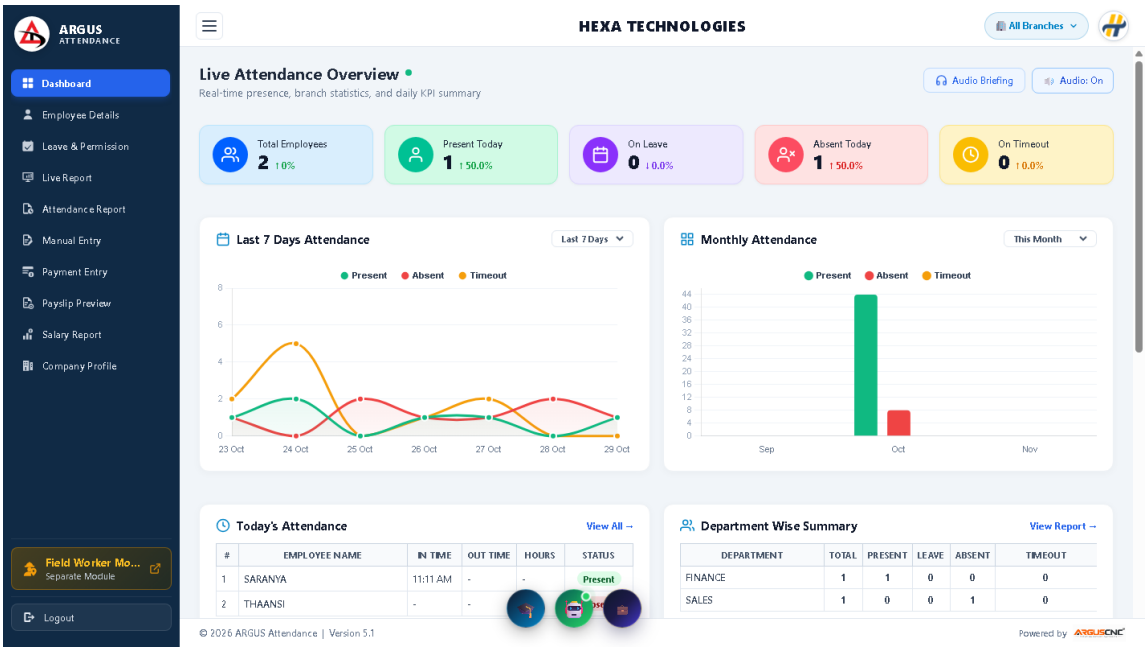


Figure 2.1: Executive Command Dashboard displaying live KPI cards, percentage rings, and quick actions

Key Dashboard Components & Operational Metrics:

Metric Card	Description & Business Value
Active Staff Count	Total number of active enrolled employees against authorized license limit.
Present Today	Real-time tally of staff who have successfully clocked in through the AI Kiosk or Mobile portal.
Late Markings	Identifies employees punching in after the configured grace period (e.g. 15 minutes past shift start).
Absent Today	Automatically computed based on shift schedules and unpunched roster records.
Overtime Active	Calculates employees continuing to work past standard shift hours for automated OT accrual.
Attendance Ring	Visual percentage representation of overall organizational presence for the active business day.
Branch Switcher	Instant navbar selector allowing management to toggle between consolidated view and individual branch offices.

SECTION 3: AI FACE RECOGNITION ATTENDANCE KIOSK

The Attendance Kiosk (/kiosk) represents the core biometric interface of the ARGUS platform. Running on any browser-equipped device (desktop terminal, dedicated wall-mounted Android tablet, or iPad), it continuously analyzes video streams to identify employees as they approach.

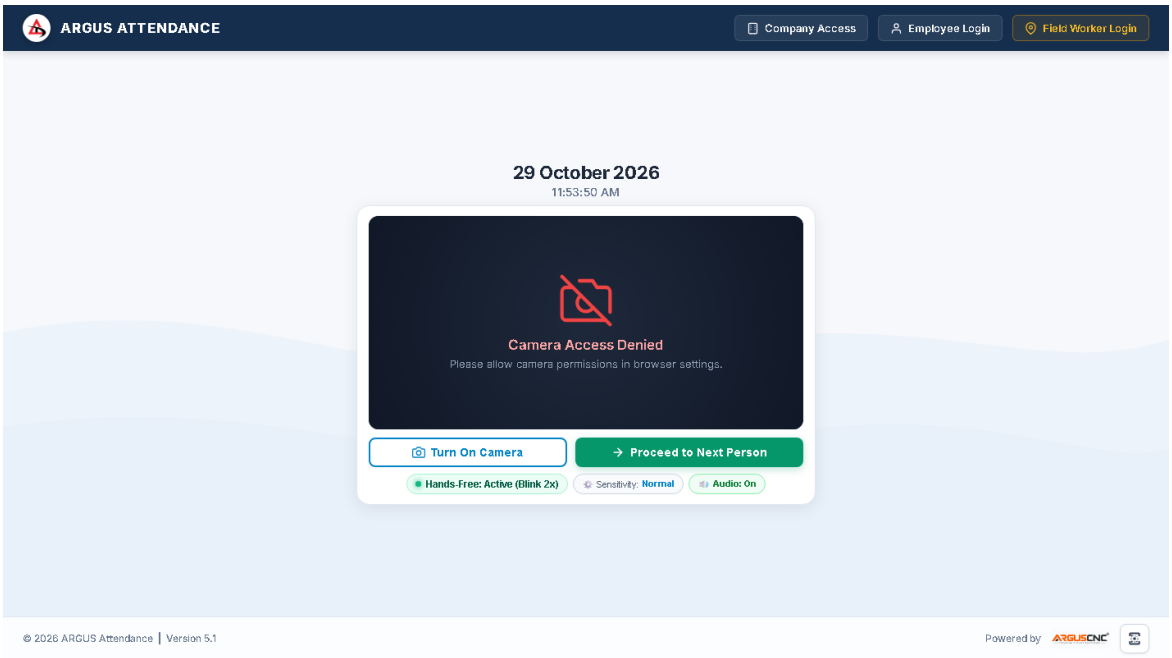


Figure 3.1: AI Face Recognition Kiosk Terminal with live camera feed, target reticle, and punch status

Biometric Engineering & Anti-Fraud Features:

- **Real-Time Face Detection & Alignment:** The neural network localizes faces within the frame, applies affine transformations for eye-nose alignment, and normalizes lighting gradients.
- **128-Dimensional Vector Embeddings:** Deep convolutional networks project facial features into a 128D mathematical space. Euclidean distance comparison identifies enrolled staff in under 400 milliseconds.
- **Anti-Spoofing & Liveness Guard:** Rejects static paper photos, tablet screen replays, and non-living representations to ensure authentic physical presence.
- **Auto-Mode Directional Intelligence:** Intelligently toggles between Punch-IN and Punch-OUT based on existing daily logs, while allowing manual mode override.
- **Audio-Visual Instant Confirmation:** Emits high-clarity synthesized chimes and displays full-screen green verification cards with employee name and exact punch timestamp.

■ Kiosk Operational Best Practice

Mount kiosk tablets at 5.0 to 5.5 feet height at entrance turnstiles or reception lobbies. Ensure indirect, consistent lighting. In environments with dynamic sunlight, ARGUS's automated contrast normalization ensures consistent 99.4% first-scan accuracy.

SECTION 4: EMPLOYEE DIRECTORY & BIOMETRIC ENROLLMENT

The Employee Details module (/employee-details) provides comprehensive human resource records management. Administrators can search, filter by branch/department, view active quota allocations, and manage onboarding.

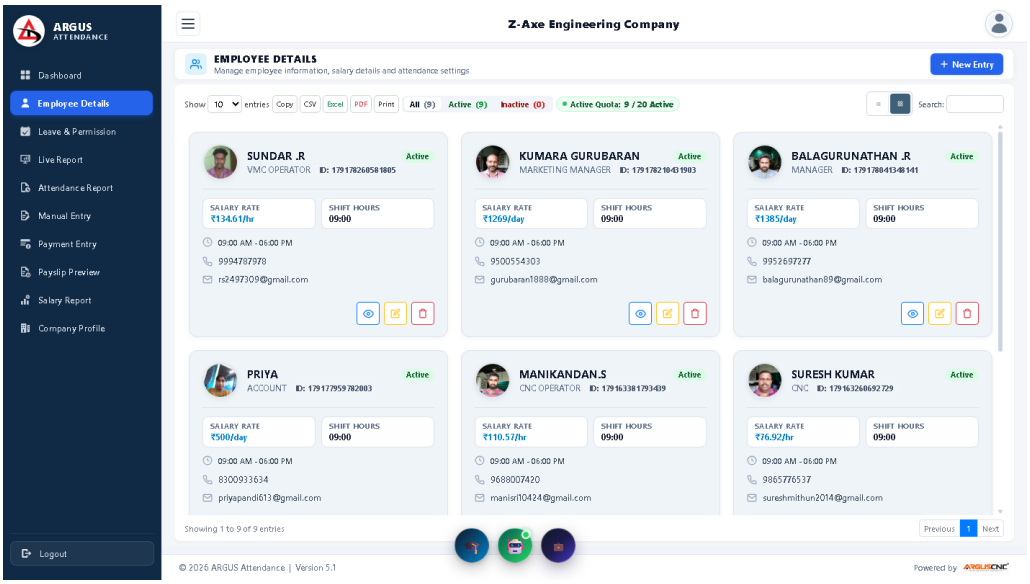


Figure 4.1: Employee Directory Roster displaying department badges, wage classification, and quick actions

Biometric Face Registration & Onboarding Modal

Clicking **'New Entry'** opens the enrollment dialog. Administrators configure employee demographics, shift assignments, and wage structures, while capturing reference facial photos via connected webcam or high-res photo upload.

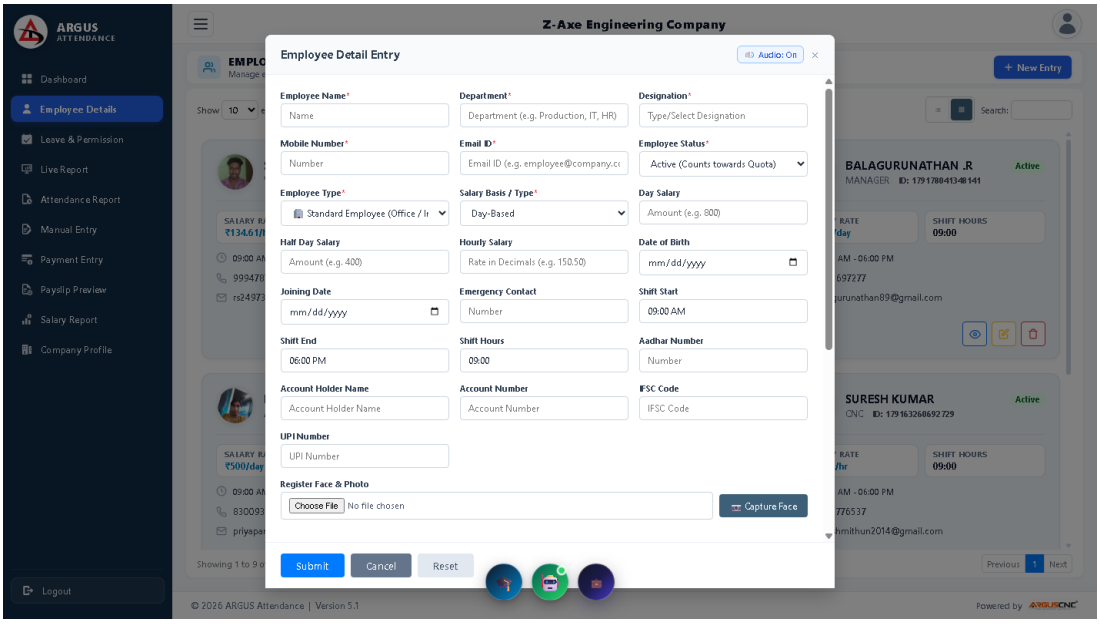


Figure 4.2: Comprehensive Employee Enrollment Dialog with wage structure, PF/ESI parameters, and face enrollment

Configurable Employee Parameters:

- **Wage Basis:** Daily Wage or Monthly Fixed Salary with automatic daily rate conversion.
- **Statutory Allowances:** Base Salary, Dearness Allowance (DA), House Rent Allowance (HRA), and Conveyance.
- **Compliance Toggles:** PF (Provident Fund) and ESI (Employee State Insurance) deduction eligibility.
- **Overtime Rule:** Configurable OT rate multiplier (e.g. 1.0x or 1.5x) or complete OT exemption.

SECTION 5: REAL-TIME MONITORING, REPORTS & AUDIT OVERRIDES

Complete transparency is provided through three complementary reporting views: the instant chronological Live Feed, the multi-filtered Historical Report, and the audit-logged Manual Entry tool.

5.1 Real-Time Live Attendance Stream

The Live Report (/live-report) is an auto-refreshing surveillance stream. Every punch immediately displays the exact timestamp, employee name, branch location, IN/OUT tag, and live camera snapshot proof.

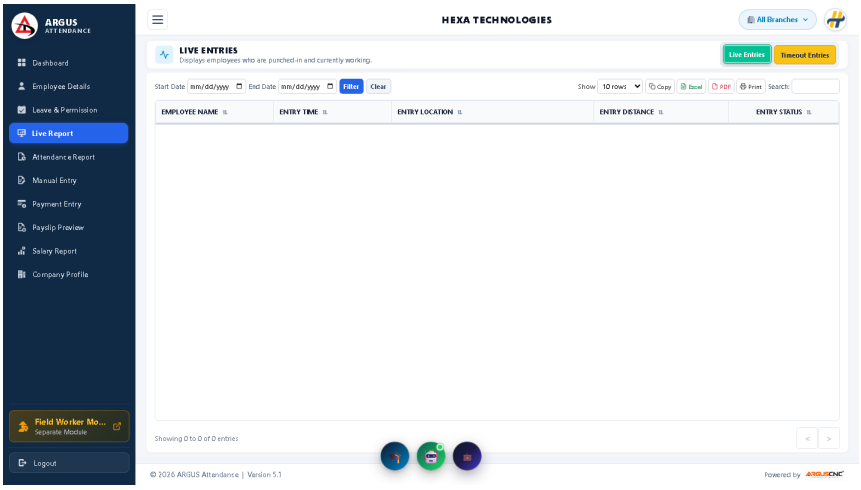


Figure 5.1: Real-Time Live Attendance Feed with camera verification captures

5.2 Historical Attendance Analytics & Reporting

The Attendance Report (/attendance-report) aggregates historical data over customizable date ranges. It analyzes total days present, half-days, late occurrences, total duty hours, and overtime accrued, with one-click export to Excel, CSV, or formatted PDF.

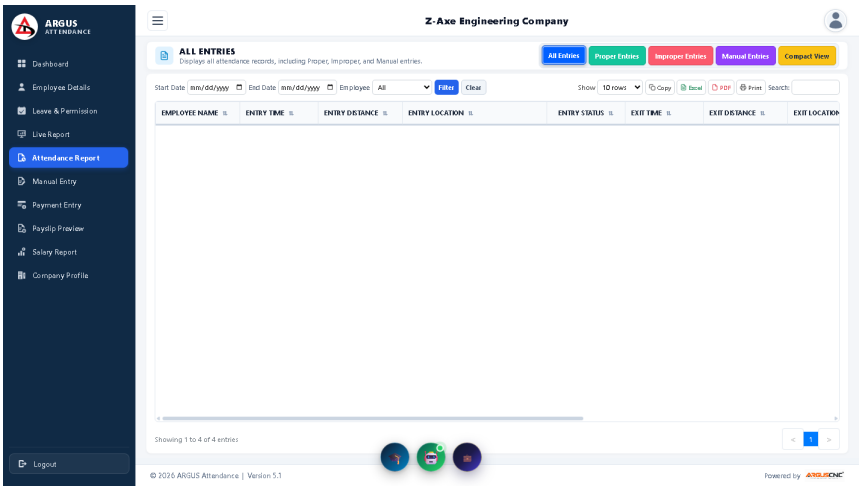


Figure 5.2: Historical Attendance Analytics with multi-parameter filtering and export toolbar

5.3 Manual Entry & Audit Adjustments

When an employee forgets to clock out or undergoes off-site assignments, authorized supervisors can log manual attendance corrections through /manual-entry. To ensure strict compliance and prevent favoritism, all manual entries require mandatory supervisory notes and are permanently flagged in audit logs.

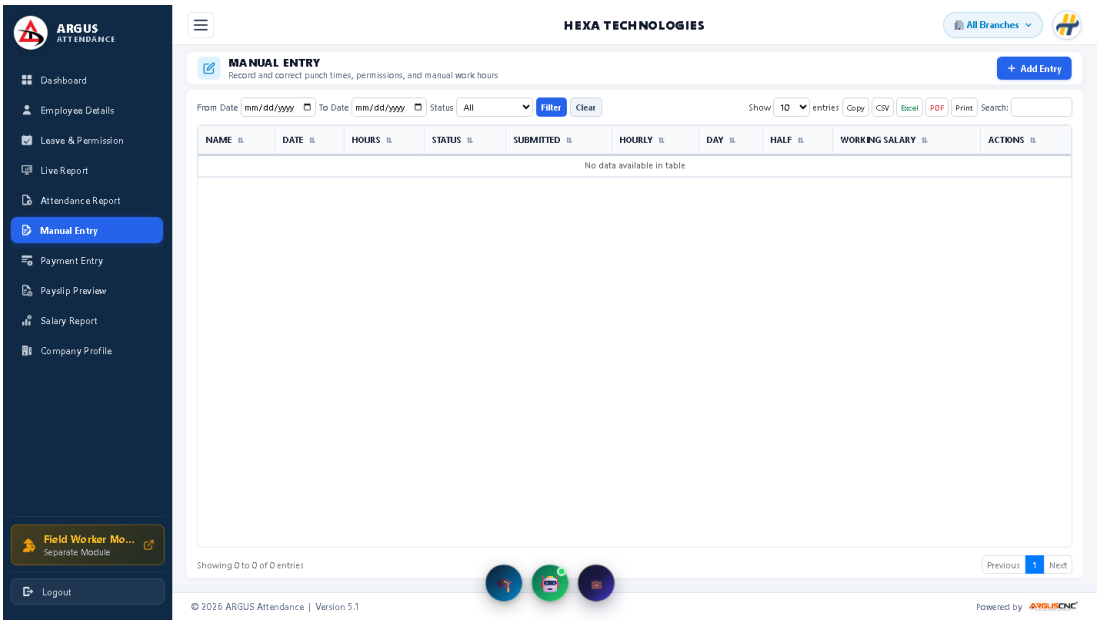


Figure 5.3: Manual Attendance Entry Screen with mandatory audit remarks

■■ Compliance & Tamper-Proof Audit Trail

Every attendance modification—whether created by AI facial detection, supervisor manual entry, or mobile check-in—records the initiating user, timestamp, client IP, and justification. This eliminates ghost employee fraud and satisfies strict ISO / labor audit requirements.

SECTION 6: LEAVE GOVERNANCE & BALANCE TRACKING

ARGUS streamlines time-off administration with an integrated Leave & Permission workflow that connects employee applications directly to supervisor review desks and payroll calculations.

6.1 Leave & Permission Review Workflow

Supervisors access /leave-permission to review pending applications. The sidebar dynamically displays real-time badge counters for unreviewed requests. Applications can be approved or rejected with written remarks in a single click.

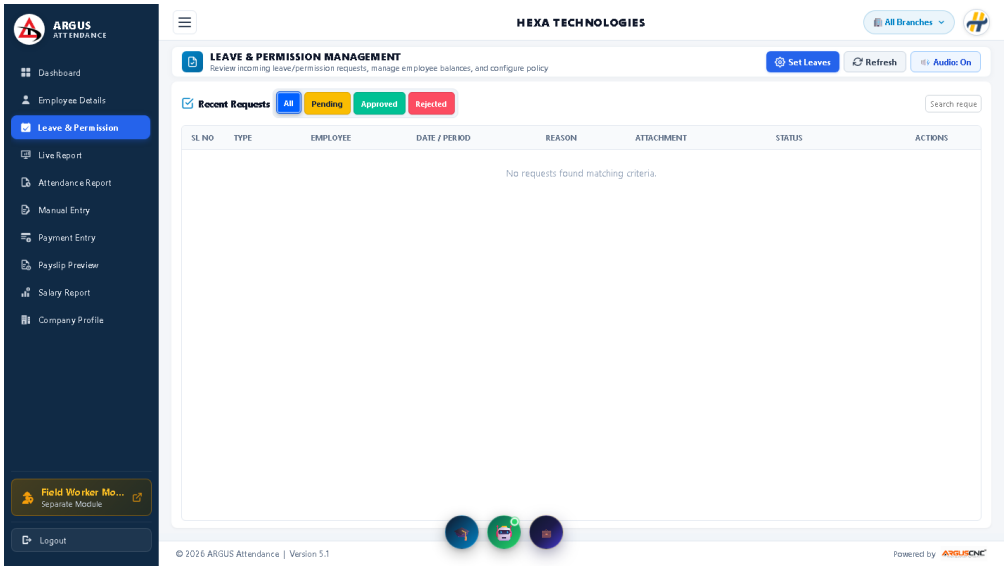


Figure 6.1: Leave & Permission Management Workspace with pending application badges

6.2 Consolidated Balance Sheet

The Balance Report (/balance-report) delivers an executive summary of outstanding employee entitlements and liabilities, comparing statutory leave balances (CL, SL, EL) against consumed days and tracking open salary advance balances.

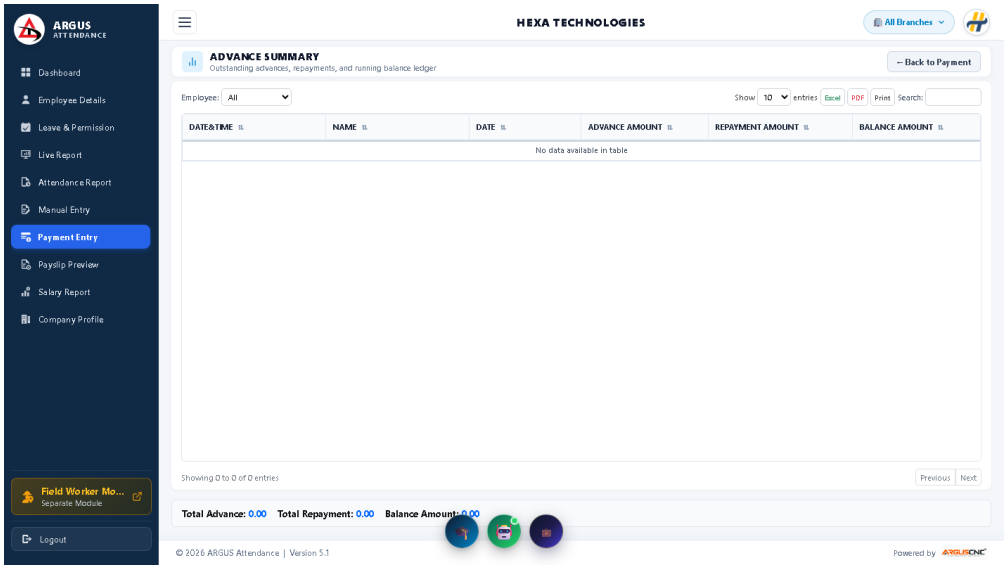


Figure 6.2: Consolidated Employee Leave & Advance Balance Report

SECTION 7: AUTOMATED PAYROLL, ADVANCES & PAYSLIPS

The ARGUS payroll module automates wage computation by linking verified attendance logs directly to salary disbursals, advance loan deductions, and statutory compliance withholdings.

7.1 Salary Advance & Loan Management

Organizations can disburse salary advances through /advance-management. The system establishes automated repayment schedules, automatically deducting installment amounts from subsequent monthly payroll runs until fully liquidated.

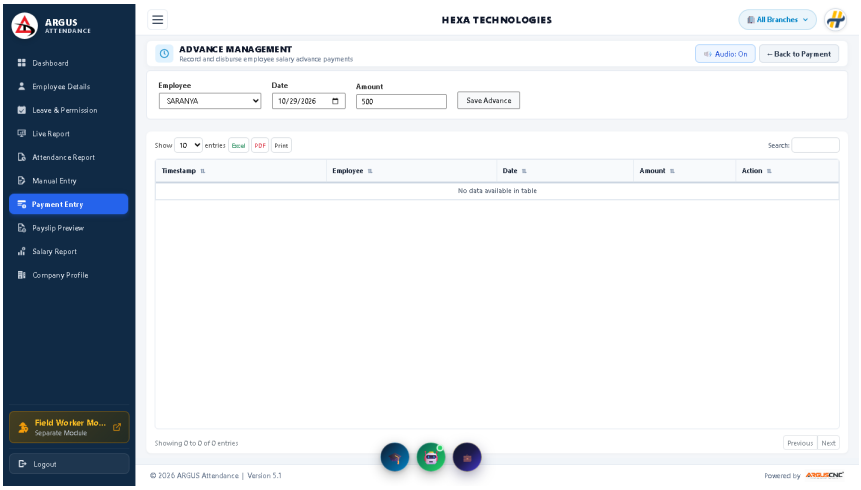


Figure 7.1: Salary Advance & Loan Installment Ledger

7.2 Payment Entry & Disbursal Vouchers

The Payment Entry module (/payment-entry) records disbursements executed via Cash, Bank Transfer (NEFT/RTGS), or UPI, automatically generating digital payment vouchers and updating employee accounts.

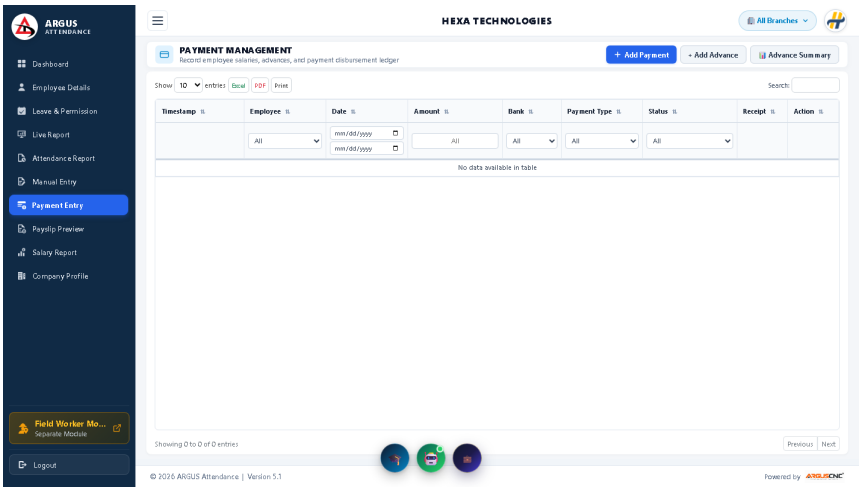


Figure 7.2: Payment Entry & Voucher Disbursal Screen

7.3 Automated Payslip Generation & Preview

Through /monthly-payslip, administrators generate audit-ready corporate payslips. The calculation engine aggregates present days, approved paid leaves, overtime hours, and allowances, while deducting PF, ESI, and advance installments to determine exact Net Pay.

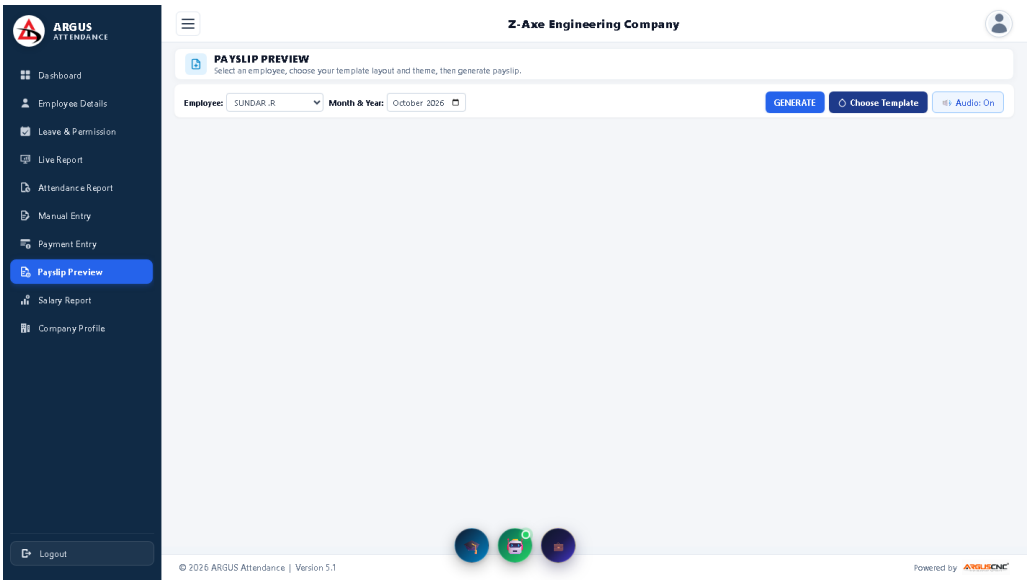


Figure 7.3: Corporate Payslip Preview with earnings, deductions, and one-click PDF export

7.4 Monthly Salary Report (Master Payroll Register)

The Salary Report (/salary-report) provides the complete monthly payroll register across the entire enterprise, summarizing total gross payout, statutory deductions, net payable salaries, and bank transfer totals.

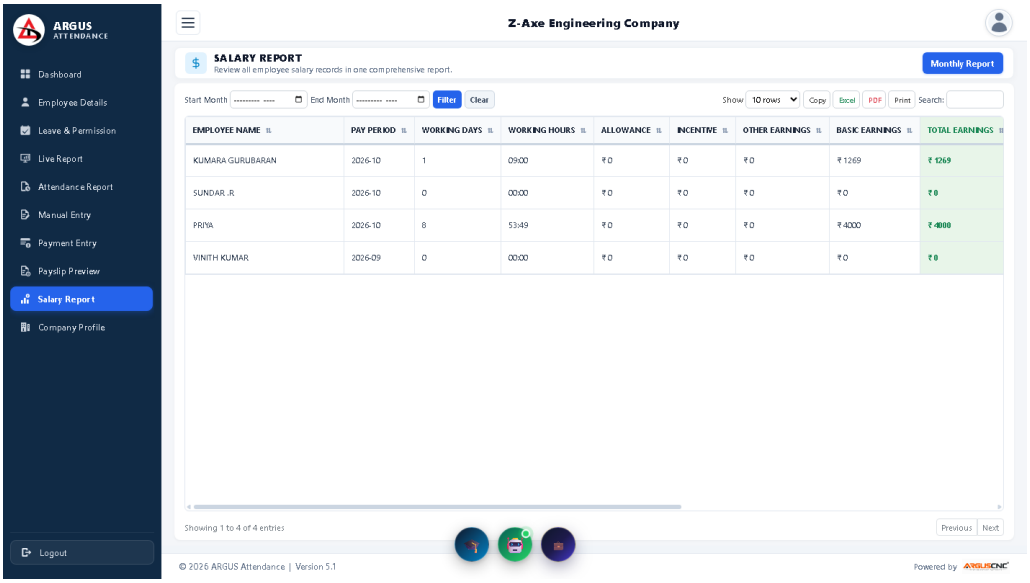


Figure 7.4: Comprehensive Monthly Salary Report & Master Payroll Register

SECTION 8: ENTERPRISE CONFIGURATION & GPS GEOFENCING

Company Profile (/company-profile) enables management to configure shifts, define strict geofence perimeters, adjust biometric sensitivity, and automate executive email report delivery.

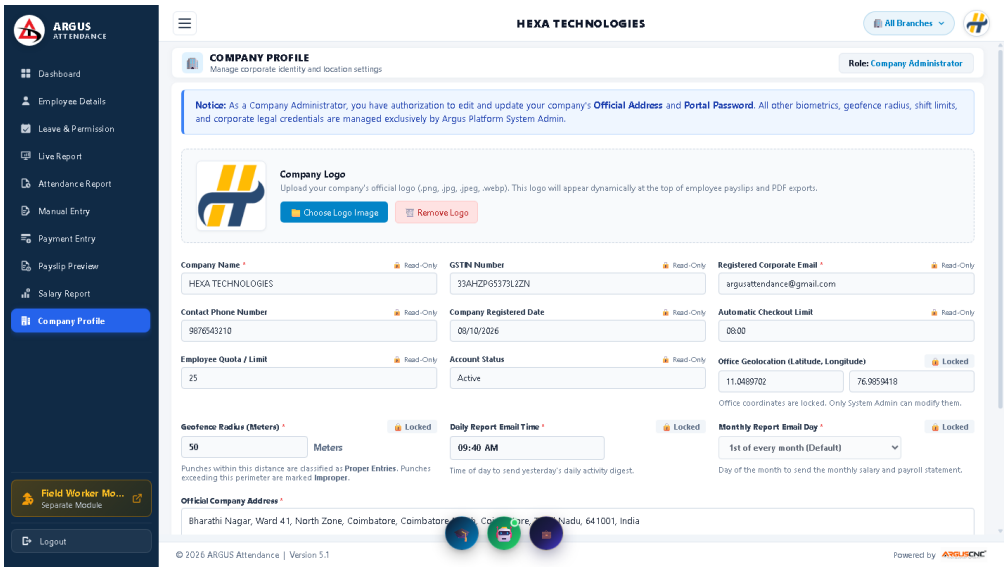


Figure 8.1: Corporate Profile Settings (GSTIN, Address, Contact Information)

Shift Scheduling, GPS Geofencing & AI Calibration

Scrolling into the configuration panel reveals granular policy parameters:

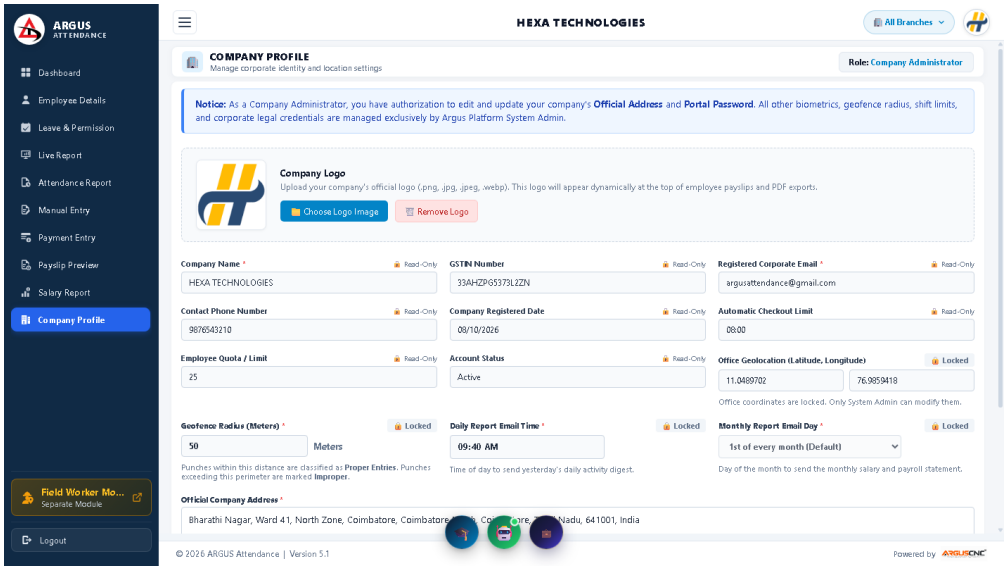


Figure 8.2: Shift Schedule, GPS Coordinate Locking, Geofence Radius, and Face Recognition Sensitivity

- **Shift Hours & Grace Period:** Defines standard start/end times and allowable arrival delay (e.g. 15 minutes) before marking late.
- **GPS Coordinate Locking:** Captures factory or office Latitude & Longitude with a tamper-resistant lock mechanism.
- **Geofence Boundary (Meters):** Enforces a strict radius (e.g. 150m to 200m) outside of which punches are automatically rejected.
- **Automated Email Reports:** Schedules daily morning digests and monthly master payroll summaries directly to executive inboxes.

SECTION 9: MULTI-BRANCH NETWORK MANAGEMENT

For multi-location manufacturing enterprises, retail chains, and corporate offices, ARGUS provides the Multi-Branch Management Module (/manage-branches).

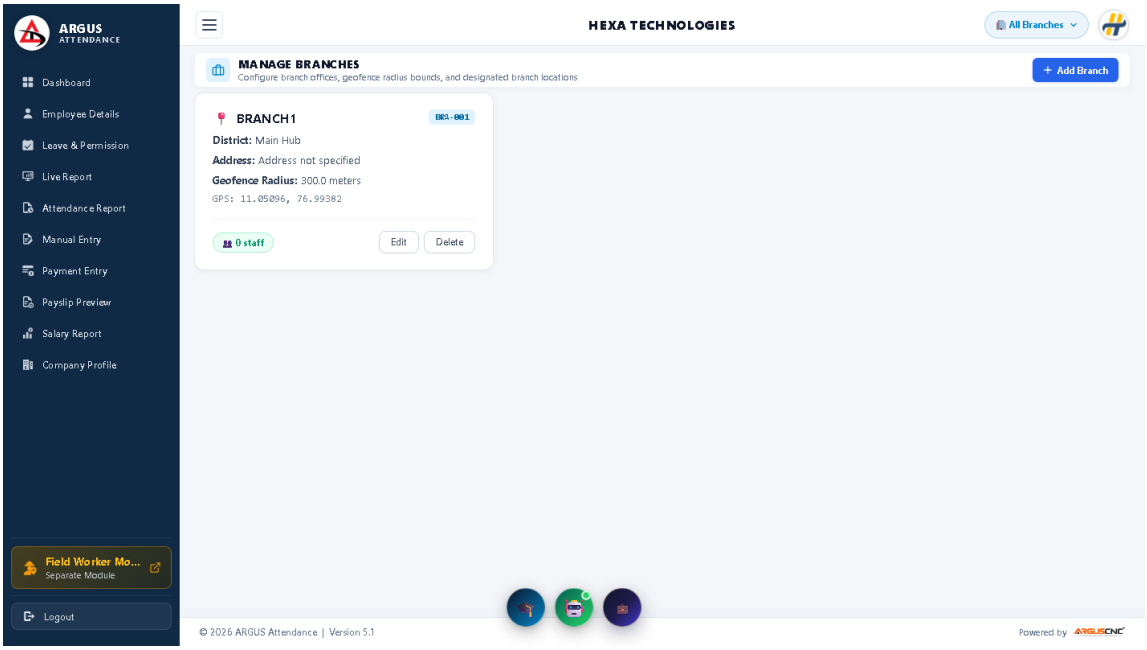


Figure 9.1: Branch Management Directory showcasing multi-hub locations and individual geofence circles

Branch Network Capabilities:

- **Independent Geofences:** Each remote branch or factory unit maintains its own GPS coordinates and independent radius.
- **Designated Branch Managers:** Assign specific managerial personnel to oversee localized attendance and leave approvals.
- **Global Top-Navbar Switcher:** Corporate leaders can effortlessly switch between 'All Branches' consolidated metrics and individual branch hubs with zero latency.
- **Branch-Filtered Reports:** Every report (Attendance, Live Stream, Salary Register) can be scoped specifically to a selected branch or viewed across the consolidated enterprise.

SECTION 10: MOBILE FIELD WORKER GPS TRACKING MODULE

Traveling sales representatives, service engineers, and logistics personnel require flexible on-the-go tracking. The Field Worker Module (/field-worker) provides GPS route tracking, visit proofs, and automated travel expense accounting.

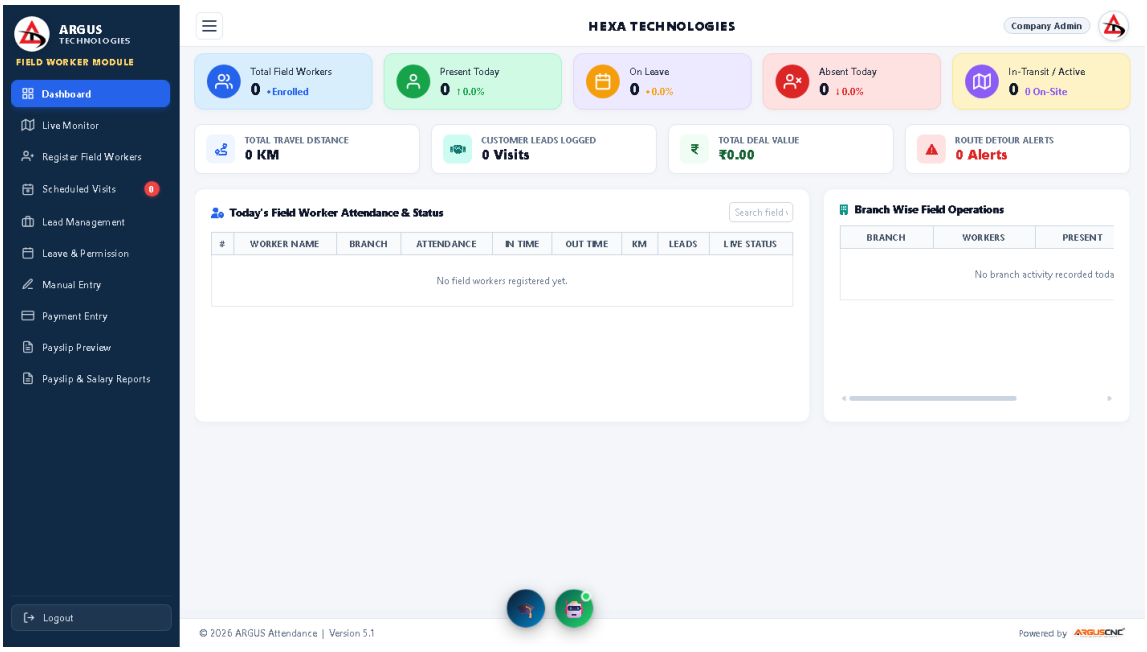


Figure 10.1: Field Worker Operational Dashboard with real-time GPS check-ins, customer visits, and map route tracking

Field Worker Capabilities & Logistics Automation:

- **Live GPS Location Check-In:** Field staff clock in from remote client sites with device GPS coordinates reverse-geocoded to real street addresses.
- **Customer Visit Logging:** Records client company name, contact person, meeting discussion summary, and order details.
- **Photo Proof of Visit:** Captures camera snapshots at the customer site to eliminate fictitious client visits.
- **Travel Route & Petrol Allowance Engine:** Computes cumulative daily kilometers traveled between visit stops and automatically calculates petrol reimbursement based on company fuel rates (e.g. ■4.50/km).

SECTION 11: SUPPORT DESK & SUPER ADMIN FLEET ADMINISTRATION

To guarantee 99.9% enterprise uptime, ARGUS embeds a dedicated Customer Support Desk for end-users and a centralized Super Admin control console for SaaS platform operators.

11.1 Customer Support Ticket Desk

Company Admins can submit technical questions, feature requests, or hardware calibration assistance directly through /support. Tickets receive immediate tracking numbers and priority classifications.

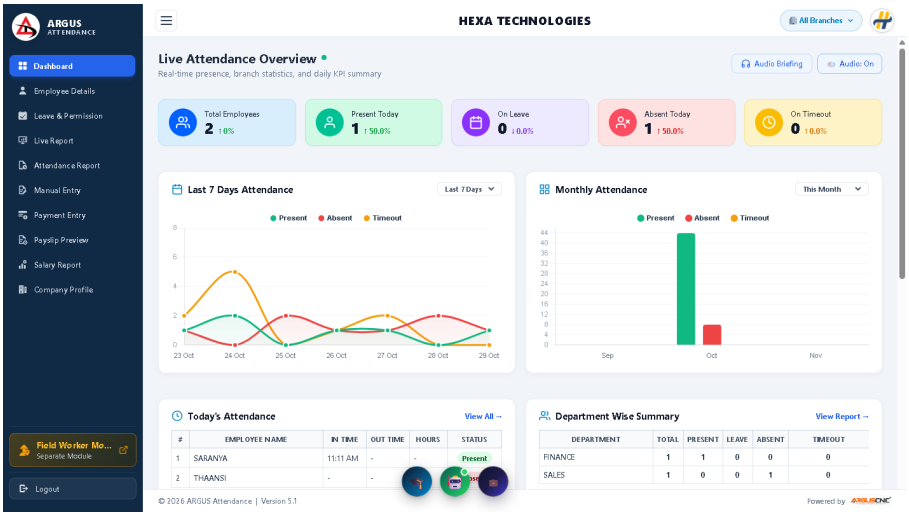


Figure 11.1: Integrated Enterprise Support Help Desk with active ticket history

11.2 Super Admin Multi-Tenant Fleet Management

Through /manage-companies, system administrators monitor client tenants, toggle modular feature packages (Single Company, Multi-Branch, Field Worker), adjust employee limits, and review platform-wide analytics.

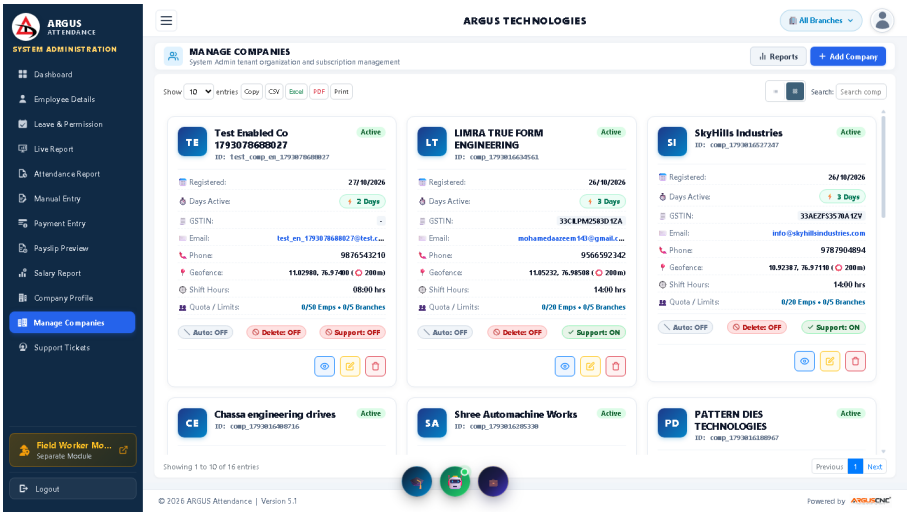


Figure 11.2: Super Admin Client Company Management Console

11.3 Super Admin Client Fleet Directory & Support Console

Super admins can inspect the full fleet of onboarded organizations, modify subscription statuses, and respond directly to customer assistance tickets.

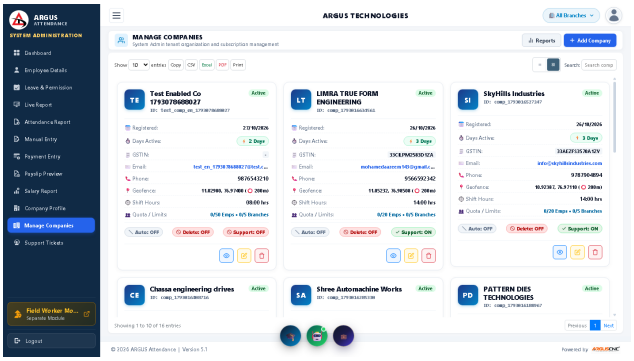


Figure 11.3: Super Admin Tenant Roster

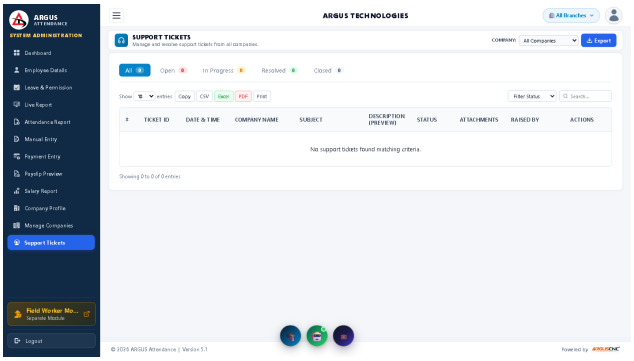


Figure 11.4: Super Admin Support Resolution Desk

SECTION 12: SECURITY SPECIFICATIONS & TECHNICAL ARCHITECTURE

The ARGUS architecture is engineered for high reliability, data privacy, and minimal maintenance across diverse enterprise environments.

Domain	Specification / Standard	Enterprise Benefit
Biometric Security	Non-reversible 128D mathematical vectors	Original face photos cannot be reconstructed from vectors, ensuring strict privacy.
Authentication	PBKDF2/SHA-256 + Google Workspace OAuth	Enterprise-grade protection against brute force and session hijacking.
Geofencing Engine	Haversine spherical trigonometry ± 5m	Prevents proxy punches from outside verified workplace perimeters.
Cross-Platform	Web, Android APK, Windows Executable (.exe)	Deploy effortlessly on existing corporate hardware without lock-in.
Database Durability	MongoDB Atlas Multi-AZ Replication	99.99% uptime with automated point-in-time backup snapshots.

ARGUS TECHNOLOGIES — ADVANCED AUTOMATION & SOFTWARE DIVISION

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